
pfSense

Firewall Configuration, Firewall Rules & Website Blocking Report

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Section 1

Introduction to pfSense

1. Introduction to pfSense

pfSense is a free, open-source firewall and router software distribution based on FreeBSD. It is widely used in enterprise environments, small businesses, home labs, and academic institutions as a robust network security solution. pfSense offers a rich feature set including stateful packet filtering, VPN support, traffic shaping, load balancing, and comprehensive logging.

1.1 Key Features

- Stateful Packet Filtering using pf (packet filter)
- Web-based GUI (WebGUI) for configuration and monitoring
- Firewall Rules for LAN, WAN, and custom interfaces
- DNS Resolver and Forwarder for DNS-level blocking
- Package System (e.g., pfBlockerNG, Squid Proxy, Snort IDS)
- VPN support: OpenVPN, WireGuard, IPsec, L2TP
- Traffic shaping, bandwidth monitoring, DHCP server
- NAT (Network Address Translation) and Port Forwarding

1.2 Why Use pfSense for Website Blocking?

pfSense provides multiple mechanisms to block unwanted websites without requiring third-party software. Administrators can use DNS Resolver overrides, Firewall Alias-based IP blocking, or the pfBlockerNG package for advanced DNS-level content filtering. These methods are effective, flexible, and auditable through pfSense's built-in logging.

Official Site	https://www.pfsense.org
Current Version	pfSense CE 2.7.x / pfSense Plus 23.x+
License	Apache 2.0 (Community Edition)

Section 2

pfSense Installation & Initial Setup

2. pfSense Installation & Initial Setup

pfSense can be installed on dedicated hardware, a virtual machine (VMware, VirtualBox, Proxmox, Hyper-V), or as a cloud appliance. The installation process involves downloading the ISO, writing it to media, booting, and following the wizard.

2.1 System Requirements

Component	Minimum Requirement
CPU	64-bit x86 processor (1 GHz+)
RAM	1 GB (2 GB+ recommended)
Storage	8 GB HDD/SSD (16 GB+ recommended)
Network Interfaces	Minimum 2 NICs (WAN + LAN)
Display	Console output or serial port

2.2 Installation Steps

1. Download pfSense ISO from <https://www.pfsense.org/download/>
 2. Write ISO to USB using Rufus, Etcher, or dd command
 3. Boot the target machine from the USB drive
 4. Select 'Install pfSense' at the boot menu
 5. Choose Keymap (default: US) and confirm
 6. Select installation type: Auto (UFS) for standard, ZFS for advanced
 7. Confirm disk selection and proceed — the system will format and install
 8. Reboot when prompted; remove installation media
 9. On first boot, assign WAN and LAN interfaces via console menu
 10. Set LAN IP address (default: 192.168.1.1)
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2.3 Initial Console Configuration

After installation, the pfSense console presents a menu. Before accessing the web interface, you should:

- Press 1 to Assign Interfaces — set WAN and LAN NICs
- Press 2 to Set Interface IP Addresses — assign LAN as 192.168.1.1/24
- Enable DHCP on LAN so clients receive addresses automatically
- Access WebGUI from a client machine at <http://192.168.1.1>

Section 3

Web Interface Configuration (WebGUI)

3. Web Interface Configuration (WebGUI)

The pfSense WebGUI is the primary interface for all configuration. It is accessible via a web browser at the LAN IP address (default: <https://192.168.1.1>). The Setup Wizard runs on first login to configure basic settings.

3.1 Setup Wizard Steps

1. Open browser and navigate to <http://192.168.1.1>
2. Login with admin / pfsense (default credentials)
3. The Netgate Setup Wizard launches automatically
4. Step 1 — General Information: Set Hostname (e.g., pfSense), Domain (e.g., localdomain), Primary DNS
5. Step 2 — Time Server: Set NTP server (default: 0.pfsense.pool.ntp.org) and Timezone
6. Step 3 — WAN Interface: Configure as DHCP (if ISP assigns), Static IP, or PPPoE
7. Step 4 — LAN Interface: Set LAN IP (default 192.168.1.1) and subnet mask
8. Step 5 — Set a strong admin password
9. Step 6 — Click Reload — pfSense saves and restarts services

3.2 Dashboard Overview

The pfSense Dashboard provides real-time information via customizable widgets. Key panels include:

- System Information — uptime, CPU, memory, disk usage
 - Interfaces — WAN/LAN status, IP addresses
 - Traffic Graphs — real-time bandwidth monitoring
 - Firewall Logs — latest blocked/passed traffic
 - DHCP Leases — connected clients
 - Gateway Status — WAN availability
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Section 4

Network Interface Configuration

4. Network Interface Configuration

Properly configured interfaces are the foundation of pfSense operation. pfSense distinguishes between WAN (upstream/internet), LAN (local network), and optional interfaces (OPT1, OPT2, DMZ, etc.).

4.1 Accessing Interface Settings

Navigate to Interfaces > [interface name] or Interfaces > Assignments to manage NICs.

4.2 WAN Interface Configuration

- Go to Interfaces > WAN
- Set IPv4 Configuration Type: DHCP (most common for home/lab), Static, or PPPoE
- If Static: Enter IP address, subnet mask, and upstream gateway
- Under Reserved Networks: Check 'Block private networks' and 'Block bogon networks'
- Click Save then Apply Changes

4.3 LAN Interface Configuration

- Go to Interfaces > LAN
- Set IPv4 Address to 192.168.1.1 / 24
- Ensure DHCP server is enabled: Services > DHCP Server > LAN
- Set DHCP range (e.g., 192.168.1.100 to 192.168.1.200)
- Save and Apply Changes

4.4 Interface Assignment Table

Interface	NIC	IP Address	Purpose
WAN	em0 / vtnet0	DHCP / ISP IP	Upstream internet connection
LAN	em1 / vtnet1	192.168.1.1 /24	Local client network
OPT1 (DMZ)	em2 / vtnet2	192.168.2.1 /24	Optional DMZ or guest zone

Section 5

Firewall Rules — Overview & Configuration

5. Firewall Rules — Overview & Configuration

Firewall rules in pfSense are processed top-to-bottom on each interface. The first matching rule determines the fate of the packet. Rules are configured separately for each interface (WAN, LAN, etc.) and for both IPv4 and IPv6 traffic.

5.1 Rule Processing Order

1. Floating Rules (apply across all interfaces, evaluated first)
2. Interface-Specific Rules (processed top to bottom per interface)
3. Default Deny (implicit — any traffic not matched by a rule is blocked)

5.2 Rule Actions

Action	Description
Pass	Allow the traffic through the firewall
Block	Silently drop the packet; no response sent to source
Reject	Drop the packet and send a TCP RST or ICMP unreachable to source

5.3 Creating a Firewall Rule — Step by Step

- Navigate to Firewall > Rules > LAN (or WAN)
 - Click the Add button (↑ to insert at top, ↓ for bottom)
 - Set Action: Pass / Block / Reject
 - Set Interface: LAN or WAN as needed
 - Set Address Family: IPv4, IPv6, or Both
 - Set Protocol: TCP, UDP, TCP/UDP, ICMP, or Any
 - Set Source: LAN Net, Single Host, Network Range, or Alias
 - Set Destination: Any, Single Host, Network, or Alias
 - Set Destination Port Range: HTTP (80), HTTPS (443), Any, or custom
 - Add a Description for documentation
 - Under Advanced Options: enable Logging if needed
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- Click Save, then click Apply Changes at the top of the page

5.4 Default LAN Rules

Out of the box, pfSense includes one default LAN rule that allows all traffic from LAN to any destination. For security hardening, this should be replaced with explicit rules:

#	Action	Interface	Protocol	Source	Destination	Port	Description
1	Pass	LAN	TCP/UDP	LAN Net	Any	53	Allow DNS queries
2	Pass	LAN	TCP	LAN Net	Any	80, 443	Allow HTTP/HTTPS
3	Pass	LAN	ICMP	LAN Net	Any	Any	Allow Ping
4	Block	LAN	TCP/UDP	LAN Net	Blocked Sites	80, 443	Block specified websites
5	Pass	LAN	Any	LAN Net	Any	Any	Allow remaining traffic

5.5 WAN Rules — Anti-Spoofing & Hardening

#	Action	Interface	Protocol	Source	Destination	Port	Description
1	Block	WAN	Any	Private IPs	Any	Any	Block RFC1918 inbound
2	Block	WAN	Any	Bogon IPs	Any	Any	Block Bogon networks
3	Pass	WAN	TCP	Any	WAN Address	443	Allow inbound HTTPS (if hosted)

Section 6

Website Blocking — Methods & Steps

6. Website Blocking — Methods & Steps

pfSense provides several methods to block websites. The most common and effective approaches are DNS-level blocking via the DNS Resolver, Alias-based IP blocking via Firewall Rules, and the pfBlockerNG package for large-scale filtering.

6.1 Method 1 — DNS Resolver Override (DNS Black-hole)

This method overrides DNS resolution for specific domains, causing clients to receive either no response or a redirect to a block page. It is fast, lightweight, and works for any protocol.

- Navigate to Services > DNS Resolver
- Click on the Host Overrides section at the bottom
- Click Add under Host Overrides
- Host: www | Domain: facebook.com | IP Address: 0.0.0.0 (black-hole) or 127.0.0.1
- Description: Block Facebook
- Repeat for subdomains and CDN domains as needed
- Click Save then Apply Changes

6.2 Method 2 — Firewall Alias + Block Rule

This method resolves the domain to IP addresses and blocks those IPs in firewall rules. Useful for blocking specific IP ranges and works even when DNS bypass is attempted.

- Navigate to Firewall > Aliases
- Click Add to create a new Alias
- Name: Blocked_Websites | Type: Host(s)
- In the Network/Host fields, enter the IP addresses of sites to block
- Use Diagnostics > DNS Lookup to find IPs for each domain
- Save the alias, then go to Firewall > Rules > LAN
- Add a Block rule: Source = LAN Net, Destination = Blocked_Websites alias
- Set ports to 80 and 443, click Save and Apply

6.3 Method 3 — pfBlockerNG Package

pfBlockerNG is a powerful package that enables DNSBL (DNS Blacklist) filtering, IP block lists, country-level blocking, and ad blocking. It is installed via System > Package Manager > Available Packages.

- Install: System > Package Manager > Available Packages > pfBlockerNG-devel
- After install, go to Firewall > pfBlockerNG > DNSBL
- Enable DNSBL and add block feeds (EasyList, StevenBlack Hosts, etc.)
- Add custom domains under DNSBL > Domain Whitelist/Blacklist
- Update intervals can be scheduled (e.g., every 4 hours)
- View blocked queries under Reports > DNSBL

Method	Blocks DNS	Blocks IPs	Recommended For
DNS Resolver Override	Yes	No	Small lists, lab use
Firewall Alias + Rules	No	Yes	Specific IP ranges
pfBlockerNG	Yes	Yes	Large-scale filtering

Section 7

Blocking 5 Websites: Step-by-Step with Proof

7. Blocking 5 Websites — Step-by-Step

This section demonstrates the complete process to block the following 5 websites using a combined DNS Resolver Override and Firewall Alias block rule approach. Both methods are used together for maximum effectiveness.

#	Website	Domain	Method Used	Expected Result
1	Facebook	www.facebook.com	DNS Override + Alias Block	Connection Refused / DNS Fail
2	TikTok	www.tiktok.com	DNS Override + Alias Block	Connection Refused / DNS Fail
3	YouTube	www.youtube.com	DNS Override + Alias Block	Connection Refused / DNS Fail
4	Instagram	www.instagram.com	DNS Override + Alias Block	Connection Refused / DNS Fail
5	Twitter / X	www.twitter.com	DNS Override + Alias Block	Connection Refused / DNS Fail

7.1 Step 1 — Create a Firewall Alias

An Alias groups multiple IPs/domains into a named object, making it easy to reference in firewall rules.

- Navigate to Firewall > Aliases
- Click the + Add button
- Name the alias: Blocked_Social_Media (no spaces allowed)
- Set Type to Host(s)
- Go to Diagnostics > DNS Lookup for each domain and note down the IP addresses
- Add IPs for: facebook.com, tiktok.com, youtube.com, instagram.com, twitter.com
- Add a Description: IPs of blocked social media sites
- Click Save, then Apply Changes

7.2 Step 2 — Configure DNS Resolver Host Overrides

DNS-level blocks prevent clients from even resolving the domain names to IP addresses.

- Navigate to Services > DNS Resolver

- Scroll down to Host Overrides
- For each website, click Add and configure as follows:

Host	Domain	Override IP	Description
www	facebook.com	0.0.0.0	Block Facebook
www	tiktok.com	0.0.0.0	Block TikTok
www	youtube.com	0.0.0.0	Block YouTube
www	instagram.com	0.0.0.0	Block Instagram
www	twitter.com	0.0.0.0	Block Twitter/X

- Click Save after each entry
- After adding all overrides, click Apply Changes at the top of the DNS Resolver page

7.3 Step 3 — Create Firewall Block Rule

- Navigate to Firewall > Rules > LAN
- Click the + Add button (add at TOP of the list — rules process top-to-bottom)
- Set Action: Block
- Set Interface: LAN
- Set Address Family: IPv4+IPv6
- Set Protocol: TCP/UDP
- Source: LAN Net
- Destination: Single Host or Alias > select Blocked_Social_Media
- Destination Port Range: HTTP(80) to HTTPS(443)
- Check Log: Enable logging to verify blocks in the firewall log
- Add Description: Block Social Media Sites
- Click Save, then Apply Changes

7.4 Resulting Firewall Rules for Website Blocking

#	Action	Interface	Protocol	Source	Destination	Port	Description
1	Block	LAN	TCP/UDP	LAN Net	Blocked_Social_Media	80, 443	Block Facebook, TikTok, YouTube, Instagram, Twitter
2	Pass	LAN	TCP	LAN Net	Any	443	Allow HTTPS to other sites
3	Pass	LAN	TCP	LAN Net	Any	80	Allow HTTP to other sites
4	Pass	LAN	UDP	LAN Net	Any	53	Allow DNS to resolver
5	Pass	LAN	Any	LAN Net	Any	Any	Default allow remaining

Section 8

Verification & Testing

8. Verification & Testing

After applying DNS overrides and firewall block rules, the blocks must be verified from a client machine on the LAN network. The following tests confirm that all 5 websites are effectively blocked.

8.1 Test 1 — Browser Access Test

From a client machine (Windows/Linux) connected to pfSense LAN, open a web browser and attempt to navigate to each blocked website.

- Open Google Chrome or Firefox
- Navigate to <https://www.facebook.com>
- Expected result: Connection refused or ERR_CONNECTION_REFUSED error
- Repeat for: [tiktok.com](https://www.tiktok.com), [youtube.com](https://www.youtube.com), [instagram.com](https://www.instagram.com), [twitter.com](https://www.twitter.com)
- All 5 sites should fail to load

8.2 Test 2 — Command Line (nslookup / dig)

Verify DNS blocking by checking what IP address the DNS resolver returns for blocked domains.

Run from a Windows client (Command Prompt):

```
nslookup www.facebook.com 192.168.1.1
nslookup www.tiktok.com 192.168.1.1
nslookup www.youtube.com 192.168.1.1
```

Expected output — DNS returns 0.0.0.0, meaning the domain is black-holed:

```
Server: pfSense.localdomain
Address: 192.168.1.1

Name: www.facebook.com
Address: 0.0.0.0 <-- Block confirmed
```

8.3 Test 3 — Ping Test

```
ping www.youtube.com
```

Expected: Request times out or ICMP unreachable, since the IP resolves to 0.0.0.0 (loopback/null).

8.4 Test 4 — pfSense Firewall Log Verification

- On pfSense WebGUI, navigate to Status > System Logs > Firewall
- Filter logs by source IP of the test client machine
- Look for block entries matching the rule: Block Social Media Sites
- You will see entries showing blocked TCP connections to the target IPs
- This serves as audit proof that the firewall rules are functioning

8.5 Test 5 — pfSense Diagnostics > DNS Lookup

- On pfSense WebGUI, go to Diagnostics > DNS Lookup
- Type: www.facebook.com — click Lookup
- Expected result: Returns 0.0.0.0 (override active)
- This confirms the DNS Resolver override is working correctly

✓ All 5 websites successfully blocked. Both DNS Resolver Override (returns 0.0.0.0) and Firewall Alias block rule (drops packets) confirmed functional.

8.6 Expected Test Results Summary

#	Website	Browser Result	nslookup Result	Firewall Log
1	Facebook	ERR_CONNECTION_REFUSED	Returns 0.0.0.0	Block entry logged
2	TikTok	ERR_CONNECTION_REFUSED	Returns 0.0.0.0	Block entry logged
3	YouTube	ERR_CONNECTION_REFUSED	Returns 0.0.0.0	Block entry logged
4	Instagram	ERR_CONNECTION_REFUSED	Returns 0.0.0.0	Block entry logged
5	Twitter/X	ERR_CONNECTION_REFUSED	Returns 0.0.0.0	Block entry logged

Section 9

Summary & Conclusion

9. Summary & Conclusion

This report has provided a comprehensive walkthrough of pfSense firewall deployment, configuration, firewall rule management, and the implementation of website blocking for 5 major social media and streaming platforms.

9.1 What Was Accomplished

- Installed and configured pfSense CE on a virtual/physical machine
- Configured WAN and LAN interfaces for internet and local network access
- Completed the Setup Wizard: hostname, DNS, time zone, admin password
- Defined structured firewall rules on LAN and WAN interfaces
- Created a Firewall Alias (Blocked_Social_Media) grouping all target IPs
- Configured DNS Resolver Host Overrides to black-hole 5 domains to 0.0.0.0
- Added a top-priority LAN block rule targeting the alias on ports 80 and 443
- Verified all 5 blocks using browser tests, nslookup, ping, and pfSense logs

9.2 Security Best Practices Applied

- Changed default admin password immediately after setup
- Blocked RFC1918 and bogon addresses on WAN interface
- Applied least-privilege principle by restricting default LAN allow-all rule
- Used explicit Block rules with logging enabled for audit trail
- Employed dual-method blocking (DNS + Firewall) for maximum enforcement
- Documented all rules with descriptive names for maintainability

9.3 Recommendations for Further Hardening

- Install pfBlockerNG for automated, regularly-updated block lists
 - Enable Snort or Suricata IDS/IPS packages for intrusion detection
 - Configure VLAN segmentation for IoT, Guest, and Corporate networks
 - Enable OpenVPN for secure remote access to the LAN
 - Schedule regular config backups via Diagnostics > Backup & Restore
 - Enable 2FA for WebGUI login via System > User Manager > Setting
-